

Game-Diffusion-based Multi-Agent Learning of Ramp Merging Coordination for Connected and Automated Vehicles

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ABSTRACT Coordinated decision-making among connected and autonomous vehicles (CAVs) during highway ramp merging presents a significant challenge in partially observable environments with mixed motives. In these scenarios, competitive interactions often compromise safety, efficiency, and passenger comfort. Conventional multi-agent reinforcement learning (MARL) approaches, such as multi-agent proximal policy optimization (MAPPO), lack theoretical optimality guarantees and often produce discontinuous control actions that degrade ride comfort and system reliability. To address these limitations, this work proposes Game-Diff MARL, a framework that integrates diffusion-based generative modeling with game-theoretic reasoning for robust and smooth CAV coordination. The methodology integrates three components. First, a diffusion refinement engine generates temporally coherent trajectories through iterative denoising. Second, a game-theoretic strategy engine refines these smooth actions into strategically optimal policies based on Nash or Stackelberg equilibria. Third, a state-adaptive gating network dynamically balances strategic optimality with motion smoothness. Evaluated on HighwayEnv, Game-Diff MARL reduces collision rates by 70% (from 12.3% to 3.7%) compared to the baselines. Theoretical analysis supports the proposed framework with convergence arguments toward an approximate Nash equilibrium and bounds on action smoothness.

INDEX TERMS Connected Autonomous Vehicles (CAVs), Diffusion Models, Game Theory, Highway Ramp Merging, Multi-Agent Reinforcement Learning (MARL)

I. INTRODUCTION

Connected and autonomous vehicles (CAVs) are driving the transformation of intelligent transportation systems (ITS), promising to elevate traffic safety, throughput, and passenger comfort [1], [2]. However, coordinated decision-making for CAVs in highway ramp merging—a classic traffic bottleneck with inherent competitive-cooperative multi-agent interactions—remains a pivotal unsolved challenge that impedes large-scale CAV deployment. The ramp merging problem presents three pivotal pain points that existing methods have failed to address simultaneously. (1) Safety: Collision

avoidance and safe inter-vehicle spacing are paramount; exploration phases in MARL and rule-based corrections in safety supervisors can induce high collision rates or abrupt maneuvers. (2) Efficiency: High merger success rate and throughput require coordinated utilization of the gaps, but suboptimal decisions trigger merger failures and congestion. (3) Comfort: Smooth, continuous acceleration and steering actions are essential for passenger experience and vehicle longevity, yet conventional MARL and game-theoretic methods often produce discontinuous control outputs that degrade ride quality. Within a merging zone of 200–300 m (Fig. 1),

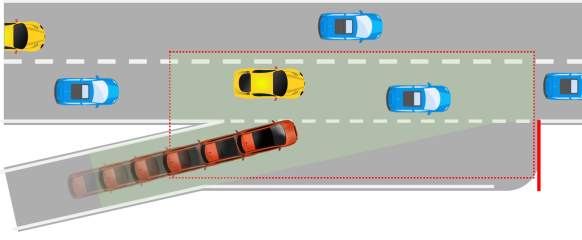


FIGURE 1: Schematic diagram of the highway ramp merging scenario which depicts the spatial layout where ramp CAVs merge into mainline traffic flow through coordinated speed adjustment and lane-merging behaviors, illustrating the competitive-cooperative interactions among vehicles.

ramp CAVs must dynamically identify valid gaps while mainline CAVs trade off travel efficiency with adaptive yield. In addition to this, CAVs rely on on-board sensors for local observations (partial observability), stochastic background traffic (mixed CAVs and human-driven vehicles, HDVs) and millisecond-level real-time requirements. Formally modeled as a Dec-POMDP, the ramp merging problem demands simultaneous optimization of safety, efficiency, and comfort—a balance that existing methods have failed to achieve effectively [3], [4].

Existing research for CAV ramp merging coordination evolves along three technical directions, each with distinct strengths and limitations. A representative safety-oriented contribution is Chen *et al.* [5], who propose a priority-based safety supervisor for highway on-ramp merging in mixed traffic. Their approach improves safety and training efficiency via multi-agent advantage actor-critic (MA2C), action masking, and curriculum learning, while intelligent driver model (IDM)/minimizing overall braking induced by lane changes (MOBIL)-based trajectory prediction is used to serially correct actions and enforce hard constraints. However, rule-based correction can induce abrupt control changes that reduce comfort, and theoretical guarantees for strategic optimality. MARL—represented by MAPPO—adopts the centralized training with decentralized execution (CTDE) paradigm and is well suited for distributed CAV control. Nevertheless, theoretical convergence guarantees to Nash equilibria are often unavailable [6], which can lead to unstable behaviors in mixed-motive scenarios. In parallel, direct policy outputs may be temporally discontinuous, and static reward weighting is often insensitive to dynamic traffic density [7], [8], [9]. Game-theoretic methods provide rigorous strategic structure but are difficult to solve online in high-dimensional continuous action spaces and may still produce abrupt control updates. Diffusion-based generative models improve temporal continuity through iterative denoising, but they do not inherently enforce equilibrium-consistent multi-agent behavior. Therefore, a critical gap remains: scalable MARL learning, game-theoretic strategic guarantees, and diffusion-level smoothness have not yet been unified in

one framework for simultaneous safety-efficiency-comfort optimization.

To bridge this gap, the ramp merging problem is formalized as an N -player stochastic game (3 agents: 2 ramp CAVs, 1 mainline CAV), and Game-Diff MARL is proposed as a unified framework that synergistically integrates diffusion-based action refinement, game-theoretic equilibrium reasoning, and state-adaptive fusion for robust, safe, efficient, and comfortable CAV ramp merging coordination. The core insight is that game theory and diffusion models offer mutually complementary strengths—game theory ensures strategic stability for multi-agent interactions, while diffusion models deliver the smooth control required for comfort—and their integration on a MARL foundation overcomes the limitations of individual approaches [10], [11]. Game-Diff MARL involves a three-stage end-to-end methodology that decouples and dynamically balances temporal smoothness and strategic optimality, featuring the following contributions: (1) a diffusion refinement engine generates temporally coherent smooth action trajectories via iterative denoising conditioned on local CAV observations, laying the foundation for ride comfort; (2) a game-theoretic strategy engine refines these smooth actions into strategically optimal policies based on Nash and Stackelberg equilibrium reasoning, aligning actions with stable game-theoretic strategies; (3) a state-adaptive gating network dynamically fuses the two action sources using real-time risk metrics, and a novel equilibrium consistency regularization term is further introduced to constrain game-theoretic refinement. Motivated by broader multi-agent coordination and safe interaction studies in robotic collaboration, this design also aligns with recent heterogeneous-agent policy optimization and interaction-aware control trends [12], [13], [14]. Prior work is cited where it most naturally supports each claim: Section II surveys MARL, diffusion-based control, game-theoretic driving, and multi-objective arbitration; Sections III and IV reference foundational definitions and training constructs as they arise; Section V links the evaluation to standard datasets, traffic models, and benchmarks. Finally, comprehensive empirical validations on HighwayEnv—a widely adopted autonomous driving simulation platform—demonstrate Game-Diff MARL’s superior performance against state-of-the-art MARL baselines.

The remainder of this paper is organized as follows. Section II reviews related work on MARL for CAV coordination, game-theoretic autonomous driving approaches, diffusion models for action generation, and safety-aware arbitration with multi-objective learning, highlighting key research gaps. Section III establishes theoretical preliminaries for MARL, game theory, diffusion models, and uncertainty-weighted multi-task learning. Section IV details the Game-Diff MARL framework, including its three-stage pipeline, core module design, and multi-objective optimization loss function. Section V presents comprehensive experimental results, including setup, performance comparisons, safety/strategic analy-

sis, ablation studies, and failure mode analysis. Section VI concludes the work and outlines future research directions.

II. RELATED WORK

Existing CAV ramp-merging research is mainly split across three lines: (i) multi-agent coordination based on MARL and game-theoretic reasoning, (ii) diffusion-based action generation for smooth control, and (iii) safety-aware arbitration and multi-objective learning that mediates competing requirements at deployment time. The discussion is kept concise while retaining representative references.

A. Multi-Agent Coordination for CAV Ramp Merging (MARL + Game Theory)

MARL methods are widely used for CAV coordination due to scalability under partial observability [3], [4]. In ramp merging, MAPPO/MA2C-style CTDE pipelines are practical, and safety-oriented designs such as Chen *et al.* [5] improve robustness via masking/supervision. MAPPO is also a strong baseline in large-scale multi-agent settings [15], [16]. Policy-gradient extensions such as MADDPG and COMA [17], [18] and factored centralized variants such as FACMAC [19] broaden the CTDE toolkit. Value-factorization methods—QMIX, VDN, and QTRAN—address credit assignment in cooperative teams [20], [21], [22]. Communication-aware MARL further targets information exchange under partial observability [23], [24]. Despite these advances, two limitations remain important for ramp merging: (1) weak equilibrium guarantees in mixed-motive interaction [6], and (2) temporally discontinuous control updates that hurt comfort. Game-theoretic approaches (Nash/Stackelberg and evolutionary formulations) address strategic structure [25], [26], but exact online equilibrium solving in continuous spaces is expensive, and equilibrium updates can still induce abrupt actions [1].

B. Diffusion-Based Action Generation in Autonomous Driving

Diffusion-based methods (e.g., Diffusion Policy and Decision Diffuser) improve temporal continuity by iterative denoising and have become promising for smooth control [11], [27]. Their robustness to multi-modal uncertainty is also attractive in traffic interaction [28], [29], and offline RL surveys highlight how expressive generative policies can leverage static datasets [30]. In autonomous driving, multi-agent diffusion variants improve trajectory realism [31], and learned traffic priors further support synthesis of stress-test driving scenarios [32]. However, most diffusion-based pipelines still focus on prediction or cooperative generation rather than equilibrium-constrained competitive-cooperative decision making. Existing work has not unified three requirements in one framework: scalable MARL learning, equilibrium-aware strategic optimization, and diffusion-level smooth control. This missing integration limits simultaneous

optimization of safety, efficiency, and comfort in CAV ramp merging. Game-Diff MARL is designed to close this gap.

C. Safety-Aware Arbitration and Multi-Objective Learning

Deployed CAV controllers must continuously balance collision avoidance, throughput, and ride comfort, particularly under sensor noise, partial observability, and mixed cooperative-competitive interactions with human-driven vehicles [6], [33]. Risk-aware cues such as time-to-collision (TTC) and gap-criticality-style indicators are standard in intelligent transportation practice for surfacing imminent conflicts, yet hard supervisory corrections or fixed priority rules can improve safety while inducing discontinuous control that conflicts with smooth, comfortable trajectories. Multi-task learning offers a complementary route by replacing static scalar reward weights with data-driven trade-offs; uncertainty-weighted objectives provide a principled way to reweight heterogeneous losses when their relative difficulty and informativeness vary across traffic regimes [34]. Despite these advances, ramp-merging pipelines seldom couple such arbitration mechanisms with scalable MARL training and generative diffusion policies: safety logic is often bolted on outside the learned policy, and equilibrium-consistent strategic refinement is rarely fused with smooth action generation in a single end-to-end framework. Game-Diff MARL is motivated by closing this gap through state-adaptive gating and joint optimization alongside diffusion and game-theoretic modules.

III. PRELIMINARIES

This section summarizes only the concepts directly used by the proposed framework, with core equations retained and lengthy derivations omitted. The goal is to provide enough technical context for the module design in Section IV while avoiding repeated background exposition.

A. Multi-Agent Reinforcement Learning

Ramp merging is modeled as a partially observable multi-agent control problem with three core CAV agents (two ramp vehicles and one mainline vehicle). Each agent receives a 25-dimensional local observation and outputs continuous actions (acceleration, steering). The joint policy value is

$$V^{\pi}(s) = \mathbb{E}_{\pi, P} \left[\sum_{t=0}^{\infty} \gamma^t \sum_{i \in \mathcal{N}} R_i(s_t, \mathbf{a}_t) \mid s_0 = s \right] \quad (1)$$

where s is the joint environment state; $\pi = (\pi_1, \dots, \pi_{|\mathcal{N}|})$ is the joint policy; P is the transition kernel; $\gamma \in (0, 1)$ is the discount factor; $\mathcal{N}$ is the agent set; $R_i(s_t, \mathbf{a}_t)$ is the reward of agent i at time t given joint action $\mathbf{a}_t$; and the expectation is over trajectories induced by π and P conditioned on $s_0 = s$.

MAPPO is used under CTDE. Its clipped objective is written as

$$L_i^{\text{CLIP}}(\theta_i) = \mathbb{E} \left[\min \left(r_i(\theta_i) \hat{A}_i, \text{clip}(r_i(\theta_i), 1 - \epsilon, 1 + \epsilon) \hat{A}_i \right) \right] \quad (2)$$

where θ_i parameterizes agent i 's policy; $r_i(\theta_i) = \pi_{\theta_i}(a | o) / \pi_{\theta_i^{\text{old}}}(a | o)$ is the importance-sampling ratio between the current and old policy (with actions a and observations o sampled on-policy); $\hat{A}_i$ is an advantage estimate for agent i ; ϵ is the PPO clip threshold; $\text{clip}(\cdot, 1 - \epsilon, 1 + \epsilon)$ clamps its argument to $[1 - \epsilon, 1 + \epsilon]$; and the expectation is over on-policy rollouts. Detailed optimization steps are omitted here and used directly in Section IV.

B. Game-Theoretic Concepts

Two notions are used in this paper. *Nash equilibrium* captures strategic stability:

$$R_i(a_i^*, a_{-i}^*) \geq R_i(a_i, a_{-i}^*), \forall i, \forall a_i \quad (3)$$

where R_i is agent i 's reward; a_i^* and a_{-i}^* are equilibrium actions for agent i and for all other agents, respectively; and a_i is any feasible deviation by agent i . *Stackelberg game* captures leader-follower priority near merge points:

$$a_L^* = \arg \max_{a_L \in \mathcal{A}_L} R_L(a_L, \text{BR}_F(a_L)) \quad (4)$$

$$\text{BR}_F(a_L) = \arg \max_{a_F \in \mathcal{A}_F} R_F(a_L, a_F) \quad (5)$$

where subscripts L and F denote leader and follower; $\mathcal{A}_L$ and $\mathcal{A}_F$ are their action spaces; R_L and R_F are their reward functions; $\text{BR}_F(a_L)$ is the follower's best response to leader action a_L ; and a_L^* is the Stackelberg equilibrium action for the leader. In this setting, the CAV nearest the merge point is treated as the leader.

C. Diffusion-Based Action Refinement

Diffusion models generate smooth actions through iterative noising and denoising [10]. In this work, denoising is conditioned on the 25-dimensional observation vector o and uses $K = 100$ diffusion steps. We write x_k for the noisy variable at step k (the same role as the noised action a^k in Section IV). A compact formulation is

$$q(x_k | x_{k-1}) = \mathcal{N}(x_k; \sqrt{\alpha_k} x_{k-1}, (1 - \alpha_k)I) \quad (6)$$

$$p_\theta(x_{k-1} | x_k) = \mathcal{N}(x_{k-1}; \mu_\theta(x_k, k), \sigma_k^2 I) \quad (7)$$

where $q(\cdot | \cdot)$ is the forward noising kernel; $p_\theta(\cdot | \cdot)$ is the parameterized reverse (denoising) model with parameters θ ; $\mathcal{N}(\cdot; \mu, \Sigma)$ denotes a Gaussian with mean μ and covariance Σ ; $\alpha_k \in (0, 1)$ is a forward-process variance schedule coefficient at step k ; I is an identity matrix of compatible dimension; μ_θ and σ_k are the reverse Gaussian mean (network output) and standard deviation at step k , respectively; and x_0 denotes a clean sample before noising.

The denoising network is trained via

$$\mathcal{L}_{\text{diff}} = \mathbb{E}_{k, x_0, \epsilon} [\|\epsilon - \epsilon_\theta(x_k, o, k)\|_2^2] \quad (8)$$

where k is uniformly drawn from $\{1, \dots, K\}$; x_0 is a clean training sample; $\epsilon \sim \mathcal{N}(0, I)$ is injected Gaussian noise; x_k is the k -step noised variable produced from (x_0, ϵ) under the forward schedule; ϵ_θ is the noise-prediction network; o is the conditioning observation; and $\|\cdot\|_2$ is the Euclidean norm.

D. Uncertainty-Weighted Multi-Task Optimization

To avoid static loss weighting, uncertainty-weighted multi-task learning is adopted [34]. The generic form is

$$\mathcal{L}_{\text{mtl}} = \sum_{m=1}^M \frac{1}{2\sigma_m^2} \mathcal{L}_m + \log \sigma_m \quad (9)$$

where $\mathcal{L}_m$ is the m -th task loss; $\sigma_m > 0$ is its learnable homoscedastic uncertainty scale; M is the number of tasks; and the summation index m is unrelated to the diffusion depth K in Section B. The instantiated objective for the proposed framework is given in Section D.

IV. METHODOLOGY

This section operationalizes the preliminaries in a unified training and inference pipeline. Fig. 2 shows the smooth-action generation engine, and Fig. 3 shows strategic refinement and action fusion for real-time ramp merging control.

A. Framework Overview

The Game-Diff MARL framework follows a three-stage inference pipeline. Stage 1 generates smooth action priors a_{diff} from local observations. Stage 2 computes equilibrium-guided strategic actions a_{game} from interaction features and game-state indicators. Stage 3 fuses a_{diff} and a_{game} into the deployed action a_{final} using the gating weight α (Eq. (18)), driven by the Game-Consistency Index (GCI, Eq. (17)) and time-to-collision (TTC, seconds). All trainable components are optimized under CTDE.

The framework is formulated as a Markov Game $\mathcal{M} = \langle \mathcal{S}, \mathcal{A}, \mathcal{P}, \mathcal{R}, \gamma, \mathcal{N} \rangle$, where $\mathcal{S}$ is the joint state space, $\mathcal{A} = \prod_{i \in \mathcal{N}} \mathcal{A}_i$ is the joint action space with per-agent sets $\mathcal{A}_i$, $\mathcal{P}(s' | s, \mathbf{a})$ is the transition kernel mapping state-action pairs $(s, \mathbf{a})$ to next states s' , $\mathcal{R} = \{R_i\}_{i \in \mathcal{N}}$ collects per-agent reward functions $R_i(s, \mathbf{a})$, $\gamma \in (0, 1)$ is the discount factor, and $\mathcal{N}$ is the agent set (3 core agents: 2 ramp, 1 mainline). A key contribution of this work is the explicit decoupling of temporal smoothness and strategic optimality: diffusion is responsible for continuity and comfort, game-theoretic reasoning is responsible for strategic correctness and safety, and gating is responsible for state-dependent arbitration. This decomposition resolves single-paradigm limitations while keeping each module's engineering role and input-output interface unambiguous.

B. Diffusion Refinement Engine

To address the action discontinuity in conventional MARL and game-theoretic methods, the Diffusion Refinement Engine generates observation-conditioned smooth actions a_{diff} for CAVs, laying the foundation for ride comfort via iterative forward noising and reverse denoising (architecture in Fig. 2). The engine consists of three core components: (1) a noise scheduler that defines the forward diffusion process, (2) a forward diffusion chain that progressively adds Gaussian noise to clean actions, and (3) a U-Net network that learns to predict the injected noise.

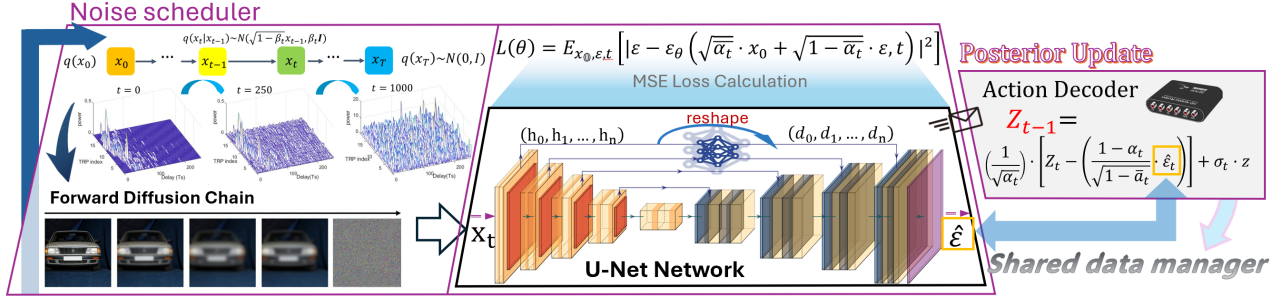


FIGURE 2: Diffusion Refinement Engine architecture. The architecture presents the noise scheduler, forward diffusion chain, and U-Net network for noise prediction. The forward process gradually injects Gaussian noise into clean actions, while the U-Net learns to predict the injected noise through MSE loss minimization, enabling iterative denoising for smooth action generation.

The forward process injects Gaussian noise into the initial action a^0 over $K = 100$ steps with a monotonically increasing noise schedule β_k , with transition and marginal distributions:

$$\begin{aligned} q(a^k | a^{k-1}) &= \mathcal{N}(a^k; \sqrt{1 - \beta_k} a^{k-1}, \beta_k \mathbf{I}) \\ q(a^k | a^0) &= \mathcal{N}(a^k; \sqrt{\bar{\alpha}_k} a^0, (1 - \bar{\alpha}_k) \mathbf{I}) \end{aligned} \quad (10)$$

where a^0 is the clean action before noising; a^k is the action corrupted after k forward steps; K is the total number of diffusion steps; $\beta_k \in (0, 1)$ is the forward variance increment at step k ; $\mathbf{I}$ is the identity matrix matching the action dimension; $\bar{\alpha}_k = \prod_{j=1}^k (1 - \beta_j)$ is the cumulative signal retention coefficient; and $\mathcal{N}(\cdot; \mu, \Sigma)$ denotes a Gaussian with mean μ and covariance Σ . The forward diffusion chain transforms clean actions into pure noise, with the marginal $q(a^K)$ approximately $\mathcal{N}(0, I)$.

The U-Net network learns to predict the noise ϵ injected during the forward process through MSE loss minimization. The training objective is:

$$\mathcal{L}_{\text{diff}} = \mathbb{E}_{k, a^0, \epsilon} [\|\epsilon - \epsilon_{\theta}(a^k, o, k)\|^2] \quad (11)$$

where $\epsilon \sim \mathcal{N}(0, I)$ is the true noise, $a^k = \sqrt{\bar{\alpha}_k} a^0 + \sqrt{1 - \bar{\alpha}_k} \epsilon$ is the noised action at step k , o is the 25-dimensional CAV local observation, and ϵ_{θ} is the U-Net noise prediction network. The U-Net architecture employs an encoder-decoder structure with skip connections, enabling effective noise prediction conditioned on observations and diffusion timesteps. The predicted noise $\hat{\epsilon} = \epsilon_{\theta}(a^k, o, k)$ is used by the reverse sampler in this engine to produce smooth action priors a_{diff} , which are then passed to the strategic refinement stage in Fig. 3.

C. Game-Theoretic Strategy Engine

The diffusion-generated a_{diff} ensures temporal smoothness but does not guarantee strategic optimality in competitive-cooperative ramp merging. As shown in Fig. 3, the Game-Theoretic Strategy Engine is dedicated to equilibrium-guided strategic refinement: it maps interaction features to Nash/Stackelberg-informed actions and outputs a_{game} . Action

fusion and final actuation are handled by the subsequent State-Adaptive Gating Mechanism, not by this engine.

1) Strategy Primitives

For each agent i , the engine refines $a_{\text{diff}, i}$ into a unified $a_{\text{game}, i}$ by fusing Nash and Stackelberg equilibrium-guided actions, derived from the agent's relative spatial-temporal features (neighboring position Δp_i , velocity Δv_i) with neighboring vehicles. The Nash equilibrium-guided action (for symmetric interaction) is:

$$a_{\text{nash}, i} = \tanh(W_p \cdot \Delta p_i + W_v \cdot \Delta v_i + b) \quad (12)$$

where $a_{\text{nash}, i} \in \mathbb{R}^{d_a}$ is the acceleration/steering action vector (d_a is the action dimension); Δp_i and Δv_i are (learned or featurized) relative position and velocity descriptors with respect to neighboring vehicles; W_p and W_v are learnable weight matrices of compatible shapes; b is a bias vector; and $\tanh(\cdot)$ applies element-wise squashing to $[-1, 1]$. The Stackelberg equilibrium (for hierarchical leader-follower merging dynamics) designates the merging-point CAV as leader L and others as followers F , with hierarchical scaling coefficients reflecting real-world traffic priority:

$$a_{\text{stack}, L} = \tanh(2W_p \cdot \Delta p_L + 2W_v \cdot \Delta v_L), \quad (13)$$

$$a_{\text{stack}, F} = \tanh(0.5W_p \cdot \Delta p_F + 0.5W_v \cdot \Delta v_F) \quad (14)$$

where Δp_L , Δv_L (resp. Δp_F , Δv_F) are the leader's (resp. follower's) relative position/velocity features, defined analogously to $\Delta p_i, \Delta v_i$ in Eq. (12); the $2\times$ scaling for leaders amplifies spatial-temporal responses, and $0.5\times$ for followers enforces conservative adjustments. The unified game-theoretic action is a weighted fusion of the two equilibria:

$$a_{\text{game}, i} = \alpha_{\text{nash}} \cdot a_{\text{nash}, i} + \alpha_{\text{stack}} \cdot a_{\text{stack}, i} \quad (15)$$

where α_{nash} and α_{stack} are learnable weights (summing to 1) that balance Nash and Stackelberg strategies, and $a_{\text{stack}, i}$ is role-dependent:

$$a_{\text{stack}, i} = \begin{cases} a_{\text{stack}, L} & \text{if agent } i \text{ is the leader} \\ a_{\text{stack}, F} & \text{if agent } i \text{ is a follower} \end{cases} \quad (16)$$

The tanh activation normalizes $a_{\text{game},i}$ to $[-1, 1]$ for stable training, and the refinement process only adjusts $a_{\text{diff},i}$ toward equilibria (rather than re-generating actions), thus preserving temporal smoothness.

2) Game-Consistency Index and Diffusion Gain Ratio

The Game-Consistency Index (GCI) is a $[0, 1]$ diagnostic metric that quantifies the multi-agent system’s strategic alignment, policy stability, and operational efficiency, defined as:

$$\text{GCI} = 1 - \frac{1}{3} (\mathcal{D}_{\text{Nash}} + (1 - \mathcal{S}) + (1 - \eta)) \quad (17)$$

where $\mathcal{D}_{\text{Nash}} = \|\pi - \text{BR}(\pi)\|_2$ is the Nash deviation, with π the joint policy, $\text{BR}(\pi)$ its best response under the current parameterization, and $\|\cdot\|_2$ the Euclidean norm; $\mathcal{S} = 1/(1 + \text{Var}(\pi))$ is policy stability, where $\text{Var}(\pi)$ denotes the temporal variance of executed actions under π ; $\eta = \sum_{i,j} R_{ij} / \sum_i \max_j R_{ij}$ is the efficiency ratio, where R_{ij} denotes the reward contribution associated with agent i under opponent index j (or the (i, j) interaction term in the centralized reward table), and the sums run over agents in $\mathcal{N}$. All components are normalized to $[0, 1]$: $\text{GCI} \rightarrow 1$ indicates optimal strategic alignment, stability, and efficiency; $\text{GCI} < 0.3$ signals severe equilibrium deviation, triggering safety interventions in the safety override mechanism (Section 3).

The Diffusion Gain Ratio (DGR) is used as an offline evaluation indicator to quantify how much reward improvement is attributable to diffusion-based smoothing. To avoid metric duplication between methodology and experiments, the formal DGR definition is reported once in the Evaluation Metrics subsection (Eq. 35).

3) Safety Override and State-Adaptive Gating

The Safety Override Mechanism provides an emergency intervention layer that prioritizes game-theoretic actions when critical safety conditions are detected. The override triggers when either of two conditions is met: (1) Time-To-Collision (TTC) < 1.5 s AND RSS (Responsive Safety System) violation, or (2) $\text{GCI} < 0.3$. When triggered, the mechanism directly outputs a_{game} as the final action, bypassing the blending module to ensure immediate collision avoidance and equilibrium recovery.

This interface exports the strategic action a_{game} and associated risk indicators (GCI, TTC, role context) to the State-Adaptive Gating Mechanism. To avoid duplicated definitions across modules, the final fusion equation and adaptive α computation are specified only in this subsection, where the final deployed action is generated. The State-Adaptive Gating Mechanism (implemented within the Blending Module of the Game-Theoretic Strategy Engine, see Fig. 3) dynamically arbitrates between a_{game} (strategic optimality/safety) and a_{diff} (temporal smoothness/comfort) based on real-time traffic conditions. The mechanism operates in coordination with the Safety Override (Section 3) to ensure robust action

selection: when safety conditions are critical, the override takes precedence; otherwise, the gating mechanism performs adaptive blending. The core fusion rule combines game-theoretic and diffusion actions:

$$a_{\text{final}} = \alpha \cdot a_{\text{game}} + (1 - \alpha) \cdot a_{\text{diff}} \quad (18)$$

where a_{final} , a_{game} , and a_{diff} are the deployed, game-refined, and diffusion-generated action vectors (same dimension d_a per agent); $\alpha \in [0.4, 0.9]$ is the adaptive fusion weight—higher values prioritize strategic stability, lower values emphasize motion smoothness. The weight is computed from the feature vector $\mathbf{x}$ (local observation o , scalar GCI, and scalar TTC in seconds), following a two-stage decision rule:

$$\alpha = \begin{cases} 0.9, & \text{if GCI} < 0.3 \text{ or TTC} < 1.5\text{s} \\ \sigma(W\mathbf{x} + b), & \text{otherwise} \end{cases} \quad (19)$$

where $\sigma(\cdot)$ is the sigmoid function; W and b are learnable weight matrix and bias mapping the gating feature vector $\mathbf{x}$ —formed by concatenating (or otherwise encoding) local observation o , GCI, and TTC—to a scalar logit before squashing. *Safety-priority mode* ($\text{GCI} < 0.3$ or $\text{TTC} < 1.5$ s): set $\alpha = 0.9$ to prioritize a_{game} . *Normal mode* ($\text{GCI} \geq 0.3$ and $\text{TTC} \geq 1.5$ s): compute α via $\sigma(W\mathbf{x} + b)$ and clip to $[0.4, 0.9]$ so both action sources remain active.

4) Social Preference and Policy Update

The framework includes social preference learning to improve cooperation and fairness. A shared data manager maintains joint information for centralized reward computation under CTDE, while social weights are predicted from joint observations via ϕ_{social} . To avoid duplicated notation, the formal socially-adjusted reward definition is provided once in Eq. 33.

The feedback layer monitors strategy entropy, GCI, and episodic performance to adapt training-time control parameters (e.g., action bounds and entropy floors) for robustness across traffic regimes. Policy updates optimize the unified objective in Section D, keeping inference logic and objective design clearly separated.

D. Multi-Objective Optimization

Multi-objective optimization for ramp merging (safety, efficiency, comfort) is formulated as a composite total loss function that integrates policy optimization, diffusion refinement, equilibrium consistency regularization, and auxiliary social preference constraints (see Fig. 3 for the complete architecture). The total loss function, implemented in the Policy Update module of the Game-Theoretic Strategy Engine, is:

$$\mathcal{L}_{\text{total}} = \frac{1}{2\sigma_1^2} \mathcal{L}_{\text{PPO}} + \frac{1}{2\sigma_2^2} \mathcal{L}_{\text{diff}} + \lambda_{\text{eq}} \mathcal{L}_{\text{consistency}} + \lambda_{\text{social}} \mathcal{L}_{\text{social}} + \log(\sigma_1 \sigma_2) \quad (20)$$

where $\mathcal{L}_{\text{PPO}}$ is MAPPO’s clipped surrogate objective (computed in the centralized training pipeline); $\mathcal{L}_{\text{diff}}$ is the diffusion reconstruction loss (expectation over diffusion step

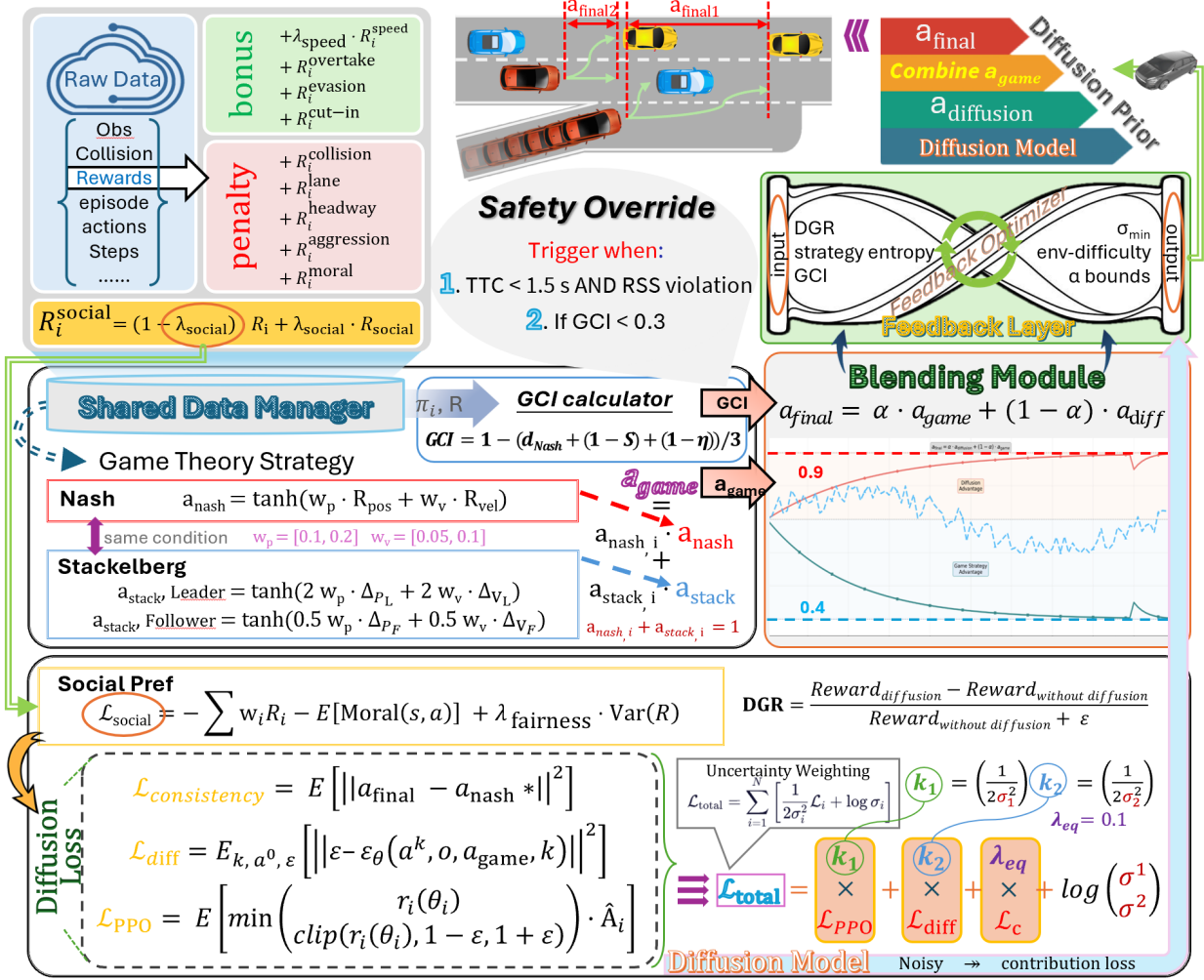


FIGURE 3: Safety-Aware Diffusion-Based Game Theory Strategy architecture. The comprehensive architecture integrates raw data processing, reward/penalty calculation, GCI computation, safety override mechanism, game-theoretic strategy engine (Nash and Stackelberg equilibria), blending module for combining game-theoretic and diffusion actions, social preference learning, feedback layer, diffusion loss components, policy update with uncertainty-weighted total loss (L_{total}), and diffusion gain ratio calculation.

k , initial action a^0 , and forward noise ϵ), trained using the U-Net network from the Diffusion Refinement Engine; $\mathcal{L}_{consistency}$ penalizes deviations of a_{final} from the Nash-guided reference a_{nash}^* (the target equilibrium action computed analogously to $a_{nash,i}$ in Eq. (12), stacked or averaged across agents as needed); λ_{eq} and λ_{social} are nonnegative scalars weighting the consistency and social terms; and $\sigma_1, \sigma_2 > 0$ are learnable uncertainty scales that adaptively balance $\mathcal{L}_{PPO}$ and $\mathcal{L}_{diff}$ following [34], with $\log(\sigma_1 \sigma_2)$ the corresponding regularizer.

Physical interpretation: A natural concern is whether game-theoretic refinement might destroy the smoothness learned by the diffusion model. The consistency term addresses this by constraining the game-theoretic output to remain within the diffusion-generated smooth action manifold—preventing “abrupt” equilibrium strategies (e.g.,

sudden braking to avoid collision) from overriding the diffusion model’s learned “driving intuition” without smooth transition. At high-risk moments (low TTC or GCI), the safety override mechanism and gating network smoothly transition toward game-theoretic actions ($\alpha \rightarrow 0.9$) rather than switching abruptly, preserving temporal coherence while prioritizing safety.

The social preference loss $\mathcal{L}_{social}$ (computed in the Social Preference module) promotes welfare maximization and fairness:

$$\mathcal{L}_{social} = - \sum_{i \in \mathcal{N}} w_i R_i - \mathbb{E} [\text{Moral}(s, a)] + \lambda_{fairness} \cdot \text{Var}(R) \quad (21)$$

where w_i are nonnegative social weights produced by ϕ_{social} (typically normalized, e.g., $\sum_i w_i = 1$); R_i is agent i ’s instantaneous or episodic reward; $\text{Moral}(s, a)$ is a scalar

ethical-preference signal depending on joint state s and joint action a ; the expectation $\mathbb{E}[\cdot]$ is over training samples; $\text{Var}(R)$ denotes the variance of $\{R_i\}_{i \in \mathcal{N}}$ across agents at a given training step (fairness regularizer); $\lambda_{\text{fairness}} = 0.02$; and $\lambda_{\text{social}} = 0.05$ scales $\mathcal{L}_{\text{social}}$ in Eq. (20) (with $\lambda_{\text{eq}} = 0.1$).

During training, reverse denoising recovers smooth actions from noisy states using the predicted noise from the U-Net, while game-theoretic guidance is injected as a strategic condition:

$$a^{k-1} = \frac{1}{\sqrt{\alpha_k}} \left(a^k - \frac{\beta_k}{\sqrt{1 - \bar{\alpha}_k}} \epsilon_{\theta}(a^k, o, a_{\text{game}}, k) \right) + \sigma_k z \quad (22)$$

where a^k and a^{k-1} are noised actions at steps k and $k-1$ of the reverse chain; β_k is the forward variance increment from Eq. (10); $\alpha_k := 1 - \beta_k$ is the per-step signal retention factor (DDPM convention); $\bar{\alpha}_k = \prod_{j=1}^k (1 - \beta_j)$ matches Eq. (10); ϵ_{θ} is the U-Net noise predictor conditioned on $(a^k, o, a_{\text{game}}, k)$, with a_{game} injecting strategic guidance; $z \sim \mathcal{N}(0, I)$ is standard Gaussian noise; σ_k is the reverse-process sampling noise scale at step k ; and I is an identity matrix matching the action dimension.

GCI (Eq. 17) and TTC are computed in real time for gating decisions and safety override triggers; social preference weights w_i are generated by a neural network ϕ_{social} from joint observations. The loss function is optimized with gradient clipping for training stability, with updates performed per episode or in batches for computational efficiency. The Feedback Layer continuously adjusts parameters ($\sigma_{\min}$, environment difficulty, action bounds) based on training metrics to maintain robust performance across diverse scenarios.

1) Analysis on Convergence

Under three realistic, standard assumptions for CAV ramp merging, the Game-Diff MARL-learned policy converges to an approximate Nash equilibrium with equilibrium-consistency regularization: (1) the reward function R_i is continuous and bounded (e.g., collision penalty ≈ -200 , speed reward ≈ 100); (2) the action space is compact (acceleration and steering within physical limits); (3) the equilibrium regularization weight $\lambda_{\text{eq}} > 0$ (set to 0.1 in this work). The $\mathcal{L}_{\text{consistency}}$ term penalizes deviations from the Nash equilibrium target and aligns denoising outputs with equilibrium signals—acting as a stabilizing bias that reduces policy exploitability in practice.

2) Analysis on Action Smoothness Bound

The diffusion refinement process provides a rigorous mathematical bound on the temporal smoothness of the action sequence $\{a_i^{(t)}\}_{t=1}^T$:

$$\|a_i^{(t)} - a_i^{(t-1)}\|_2 = O\left(\frac{\beta_{\max}}{\sqrt{K}}\right) \quad (23)$$

where $a_i^{(t)}$ is agent i 's executed action at discrete time t ; T is the episode horizon; $\beta_{\max} = \max_{1 \leq k \leq K} \beta_k$ is

Algorithm 1 Game-Diff MARL Training Algorithm.

Input: Environment env with $N = |\mathcal{N}|$ CAVs, episode horizon T , training episodes N_{ep}

Output: Trained policy π_{θ}

- 1: Initialize model parameters and on-policy rollout buffer
- 2: **for** episode = 1 to N_{ep} **do**
- 3: $s_0 \leftarrow \text{env.reset}()$
- 4: **for** $t = 0$ to T **do**
- 5: Get local observations $\{o_{i,t}\}_{i \in \mathcal{N}}$ and joint observation $\mathbf{o}_t$
- 6: Compute social preference weights $\mathbf{w}_t \leftarrow \text{softmax}(\phi_{\text{social}}(\mathbf{o}_t))$
- 7: **for** each agent $i \in \mathcal{N}$ **do**
- 8: Compute GCI_t (Eq. 17) and TTC_t (relative distance/velocity) from current state
- 9: $a_{\text{diff},i} \leftarrow \text{DiffusionSample}(o_{i,t})$
- 10: $a_{\text{game},i} \leftarrow \text{GameTheoryStrategy}(o_{i,t}, \mathbf{o}_t, a_{\text{diff},i})$
- 11: $\alpha_i \leftarrow \phi(o_{i,t}, \text{GCI}_t, \text{TTC}_t)$ with learned gating map ϕ (Eq. 19)
- 12: $a_{i,t} \leftarrow \alpha_i \cdot a_{\text{game},i} + (1 - \alpha_i) \cdot a_{\text{diff},i}$ (Eq. 18)
- 13: $a_{i,t} \leftarrow \text{Clip}(a_{i,t}, [-5, 5], [-0.5, 0.5])$
- 14: **end for**
- 15: $\mathbf{a}_t \leftarrow [a_{1,t}, \dots, a_{N,t}]$; $s_{t+1}, \mathbf{r}_t, \text{done} \leftarrow \text{env.step}(\mathbf{a}_t)$
- 16: Compute socially-adjusted rewards R_i^{social} (Eq. 33)
- 17: Store transition $(\mathbf{o}_t, \mathbf{a}_t, \mathbf{R}_t^{\text{social}}, s_{t+1})$ in rollout buffer
- 18: **if** done **then**
- 19: **break**
- 20: **end if**
- 21: **end for**
- 22: Update MAPPO actor-critic and auxiliary modules by optimizing $\mathcal{L}_{\text{total}}$ (Eq. 20)
- 23: **end for**
- 24:
- 25: **return** π_{θ}

the maximum forward-schedule increment over $K = 100$ diffusion steps; and $O(\cdot)$ denotes asymptotic order in K (informal scaling consistent with iterative denoising noise averaging). The magnitude of the action change decreases at a rate of $1/\sqrt{K}$ with more diffusion steps, motivating the observed improvements in passenger comfort.

3) Analysis on Sample Complexity Analysis

Game-Diff MARL's convergence to an ϵ -approximate Nash equilibrium follows the sample complexity bound:

$$N_{\text{samples}} = \mathcal{O}\left(\frac{1}{\epsilon^2} \log \frac{1}{\delta}\right) \quad (24)$$

where N_{samples} is the number of training samples (transitions or episodes, depending on implementation); $\epsilon > 0$ is the target accuracy for an ϵ -approximate Nash equilibrium; $\delta \in (0, 1)$ is the failure probability; and $\mathcal{O}(\cdot)$ hides constants

depending on the MDP and function classes. This bound quantifies the training samples needed to achieve an ϵ -approximate Nash equilibrium with probability at least $1 - \delta$. The quadratic dependence on $1/\epsilon$ reflects the core equilibrium learning challenge (halving ϵ requires quadrupling samples), while the logarithmic dependence on $1/\delta$ means higher confidence only needs modest sample increases. This bound confirms Game-Diff MARL retains standard MARL's sample efficiency while adding equilibrium convergence and action smoothness guarantees.

V. EXPERIMENTAL RESULTS

This section presents comprehensive empirical validations of the proposed Game-Diff MARL framework on HighwayEnv, a widely adopted autonomous driving simulation platform for ramp merging scenarios. Performance is evaluated across the three core objectives of CAV coordination—safety, efficiency, and passenger comfort—via baseline comparisons, safety evolution analysis, strategic intelligence assessment, architectural ablation studies, and failure mode analysis. All experiments adhere to the training pipeline outlined in Algorithm 1, with statistical significance established across five random seeds for all quantitative results.

A. Experimental Setup

1) Environment Configuration

HighwayEnv is configured to mimic real-world highway ramp merging dynamics, drawing inspiration from interactive driving datasets [35], [36] and motion planning benchmarks [37] that capture the adversarial and cooperative interactions inherent in merging processes. The merging scenario consists of four segments aligned with Chen *et al.* [5]: AB-1 (220 m), AB-2 (100 m), BC (100 m), and CD (100 m). This paper studies *single main lane* vehicle merging: the main highway comprises one lane into which ramp vehicles merge. Fig. 4 illustrates the single main lane layout with vehicle spawn points on the main highway and on-ramp; the merging end marks where all vehicles must complete merging. Traffic flow is modeled via the Intelligent Driver Model (IDM) [38] to ensure realistic background vehicle behavior. Stochasticity is introduced via variable counts of human-driven vehicles (HDVs) and perturbed initial vehicle speeds to enforce policy generalization across diverse traffic conditions. Key simulation parameters are summarized in Table 1, including a 200 m merging zone (consistent with real-world 200–300 m ramp merging constraints), a 25-dimensional observation space encoding relative traffic states, and a continuous action space for CAV acceleration/steering control that aligns with physical vehicle capabilities [2]. The simulation includes 3 CAVs (2 ramp, 1 mainline) and 2 background vehicles, with episodes lasting 300 steps (≈ 30 s) to ensure complete merging processes.

TABLE 1: HighwayEnv Configuration for Ramp Merging (default scenario).

Parameter	Value
Number of CAVs	3 (2 ramp, 1 mainline)
Background vehicles	2
Main lane count	1
Highway length	1000 m
Ramp length	200 m
Merging zone	200 m
Segment layout	AB-1: 220 m, AB-2/BC/CD: 100 m each
Speed limits	Mainline: 25 m/s, Ramp: 15 m/s
Episode length	300 steps (≈ 30 s)
Time step	0.1 s
Traffic model	IDM
Observation space	25-dimensional (relative states)
Action space	Continuous (acceleration, steering)
Acceleration range	$[-5, 5]$ m/s ²
Steering range	$[-0.5, 0.5]$ rad
Safe distance threshold	10 m
Safe headway threshold	1.2 s

2) Hardware and Software Environment

All experiments are conducted on a standardized test platform to ensure reproducibility. The primary hardware configuration comprises an Intel Core i9-13900K CPU (3.0 GHz, 24 cores) and an NVIDIA RTX 4090 GPU (24 GB VRAM). The software stack includes Python 3.9, PyTorch 2.0, and HighwayEnv 1.8. To validate deployment feasibility across diverse hardware, inference latency is additionally reported on an NVIDIA RTX 4060 GPU (12 GB VRAM), representative of mid-tier in-vehicle computing platforms. The test environment is documented for reproducibility; module-wise timing and cross-hardware comparison are detailed in Section 3.

3) Reward Function Design

The reward function balances three core objectives: efficiency, safety, and passenger comfort. As a theoretical framework, the reward can be conceptualized as a weighted combination of these objectives:

$$R_i^{\text{theoretical}} = 0.4 \cdot R_i^{\text{speed}} + 0.4 \cdot R_i^{\text{safety}} + 0.2 \cdot R_i^{\text{comfort}} \quad (25)$$

where v_i is agent i 's longitudinal speed and $v_{\max}$ is the scenario speed limit (Table 1); $d_{\min,i}$ is the minimum distance from agent i to any other vehicle; $d_{\text{safe}} = 10$ m is the safe-distance scale; $\Delta a_i = a_i^{(t)} - a_i^{(t-1)}$ is the acceleration increment between consecutive control steps; $R_i^{\text{speed}} = \min\left(\frac{v_i}{v_{\max}}, 1\right)$ denotes the normalized speed reward saturating at unity; $R_i^{\text{safety}} = -\exp\left(-\frac{d_{\min,i}}{d_{\text{safe}}}\right)$ is the safety penalty increasing sharply as $d_{\min,i} \rightarrow 0$; and $R_i^{\text{comfort}} = -\|\Delta a_i\|_2^2$ penalizes jerk-like acceleration changes.

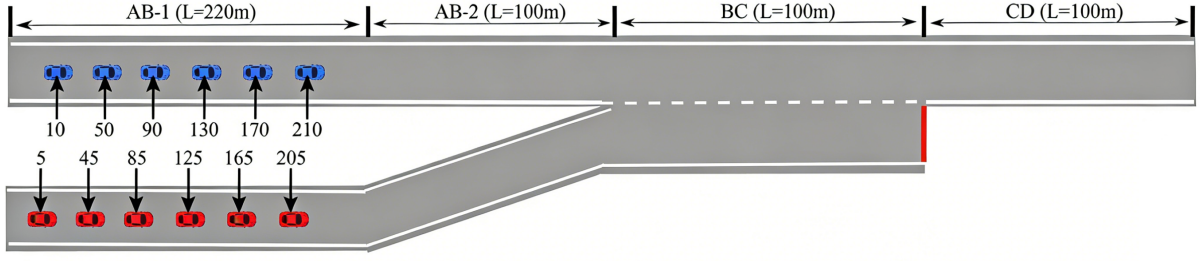


FIGURE 4: Single main lane highway ramp merging scenario. The diagram shows the segment layout (AB-1, AB-2, BC, CD) with vehicle spawn points on the single main highway lane and on-ramp, aligned with Chen *et al.* [5].

The experimental implementation employs a comprehensive reward structure that decomposes the reward into base rewards, behavioral bonuses, and penalty terms, providing fine-grained feedback for diverse traffic behaviors:

$$R_i = R_{\text{base}} + R_i^{\text{bonus}} - R_i^{\text{penalty}} \quad (26)$$

where $R_{\text{base}} = 10.0$ denotes a constant base reward that ensures non-negative total rewards, R_i^{bonus} represents the aggregated positive behavioral incentives rewarding desirable driving behaviors, and R_i^{penalty} indicates the accumulated safety and comfort penalties that penalize violations and aggressive maneuvers. The final reward is clipped to be non-negative: $R_i = \max(0, R_i)$. The comprehensive structure extends the theoretical framework by decomposing rewards into detailed components: the speed component R_i^{speed} maps directly to the speed bonus, the safety component R_i^{safety} is realized through collision and headway penalties, and the comfort component R_i^{comfort} is enforced via aggression penalties and moral constraints.

The bonus component R_i^{bonus} includes efficiency and cooperative behavior rewards:

$$R_i^{\text{bonus}} = \lambda_{\text{speed}} \cdot R_i^{\text{speed}} + R_i^{\text{overtake}} + R_i^{\text{evasion}} + R_i^{\text{cut-in}} \quad (27)$$

where $\lambda_{\text{speed}} = 1.0$ scales the efficiency component, R_i^{overtake} and R_i^{evasion} provide bonuses for overtaking and collision avoidance, and $R_i^{\text{cut-in}}$ is a context-dependent reward for lane-changing:

$$R_i^{\text{cut-in}} = \begin{cases} 10.0 \cdot \left(1 - \frac{d-5}{10}\right) & \text{if } \mathcal{C}_1 \\ -3.0 & \text{if } \mathcal{C}_2 \\ 0 & \text{otherwise} \end{cases} \quad (28)$$

where $\mathcal{C}_1$ denotes favorable cut-in conditions (moderate distance, sufficient gap, positive speed gain), $\mathcal{C}_2$ represents risky conditions (insufficient gap or marginal speed gain), d is the longitudinal distance to the target vehicle in the cut-in context, and Δv is the projected speed gain after completing the lane change (used implicitly in the predicates defining $\mathcal{C}_1$ and $\mathcal{C}_2$).

The penalty component R_i^{penalty} aggregates safety violations, lane costs, and aggressive behavior penalties:

$$R_i^{\text{penalty}} = R_i^{\text{collision}} + R_i^{\text{lane}} + R_i^{\text{headway}} + R_i^{\text{aggression}} + R_i^{\text{moral}} \quad (29)$$

where $R_i^{\text{collision}}$ is a severe penalty for collision events (CAV-to-HDV penalized more than CAV-to-CAV), and R_i^{lane} penalizes prolonged stays on merging lanes:

$$R_i^{\text{lane}} = \begin{cases} -4.0 \cdot \exp\left(-\frac{(x_i - x_{\text{merge}})^2}{10 \cdot L_{\text{merge}}}\right) & \text{if on merging lane} \\ 0 & \text{otherwise} \end{cases} \quad (30)$$

where x_i is agent i 's longitudinal position, x_{merge} the merging point, and L_{merge} the merging lane length. The headway cost R_i^{headway} penalizes insufficient following distance:

$$R_i^{\text{headway}} = \begin{cases} -4.0 \cdot \log\left(\frac{d_{\text{headway}}}{1.2 \cdot v_i}\right) & \text{if } d_{\text{headway}} < 1.2 \cdot v_i \\ 0 & \text{otherwise} \end{cases} \quad (31)$$

where d_{headway} is the bumper-to-bumper distance to the preceding vehicle, v_i is agent i 's speed, and the product $1.2 \cdot v_i$ implements a 1.2 s time-headway threshold (consistent with Table 1). The aggression penalty $R_i^{\text{aggression}}$ discourages aggressive maneuvers:

$$R_i^{\text{aggression}} = 0.5 \cdot |a_i| + 0.3 \cdot |\delta_i| + 5.0 \cdot \mathbb{I}_{\text{frequent_aggression}} \quad (32)$$

with $|a_i|$ the magnitude of agent i 's commanded acceleration, $|\delta_i|$ the magnitude of steering command, and $\mathbb{I}_{\text{frequent_aggression}} \in \{0, 1\}$ an indicator that equals 1 when a heuristic detects repeated aggressive maneuvers within a short window (and 0 otherwise). The moral penalty R_i^{moral} penalizes ethically unacceptable behaviors (intentional collisions, dangerous overtaking).

To incorporate social preferences and promote cooperative behavior, the individual reward R_i is augmented with a social welfare component that considers the collective well-being of all agents. The socially-adjusted reward is defined as:

$$R_i^{\text{social}} = (1 - \lambda_{\text{rw}}) \cdot R_i + \lambda_{\text{rw}} \cdot R_{\text{social}} \quad (33)$$

where R_i is agent i 's base reward from Eq. (26); $\lambda_{\text{rw}} \in [0, 1]$ interpolates between selfish return and team welfare (independent of the loss multiplier λ_{social} on $\mathcal{L}_{\text{social}}$ in Eq. (20)). The social welfare reward and its learnable weights are

$$R_{\text{social}} = \sum_{j \in \mathcal{N}} w_j \cdot R_j \quad (34)$$

$$\mathbf{w} = \text{softmax}(\phi_{\text{social}}(\mathbf{o}))$$

where j indexes agents in $\mathcal{N}$; R_j is agent j 's base reward (Eq. (26)); $\mathbf{w} = [w_1, \dots, w_{|\mathcal{N}|}]^\top$ is the softmax-normalized weight vector output by the social-preference network ϕ_{social} applied to the joint observation $\mathbf{o}$ (concatenation/stacking of per-agent observations under CTDE). This social preference mechanism enables agents to learn adaptive weighting strategies that balance individual and collective objectives, promoting fairness and cooperation in multi-agent interactions.

The comprehensive reward structure ensures non-negative rewards through the base reward term, provides fine-grained feedback for diverse traffic behaviors through behavioral bonuses, and maintains safety and comfort through multi-faceted penalty terms. This design enables effective learning in complex multi-agent ramp merging scenarios while preserving interpretability and alignment with real-world traffic objectives.

4) Baseline Algorithms

State-of-the-art and engineering-improved MARL algorithms are selected for performance comparison, with identical network architectures and hyperparameters to ensure fair evaluation: 3-layer MLP (128, 256, 128), Adam optimizer with $\text{lr}=3 \times 10^{-4}$, batch size=256, training steps=1M, discount factor $\gamma = 0.99$, Generalized Advantage Estimation (GAE) $\lambda = 0.95$, and PPO clipping threshold $\epsilon = 0.2$. The baselines are:

Standard MARL:

- MAPPO: Multi-agent proximal policy optimization (CTDE paradigm, shared policy, centralized critic)—the primary baseline for its strong empirical performance in cooperative traffic tasks.
- MAACKTR: Multi-agent actor-critic with Kronecker-factored trust region, with inter-agent communication (10-dimensional observation features transmitted every 5 timesteps) to mitigate environmental non-stationarity.
- MADQN: Multi-agent deep Q-network, a value-based method with discretized action space (7 acceleration levels, 5 steering angle levels)—representative of discrete control approaches for continuous CAV tasks.

Engineering-improved MARL (aligned with Chen et al. [5]):

- MAPPO+AM: MAPPO with action masking to exclude infeasible actions (e.g., collision-prone maneuvers).
- MAPPO+SS: MAPPO with rule-based safety supervisor (IDM/MOBIL trajectory prediction, serial action correction).
- Deep MARL: MAPPO with both action masking and safety supervisor—the full Chen et al. framework.

For MADQN, the continuous action space is discretized into 7 acceleration levels ($[-5, -3, -1, 0, 1, 3, 5]$ m/s²) and 5 steering angle levels ($[-0.5, -0.25, 0, 0.25, 0.5]$ rad).

5) Evaluation Metrics

Six task-aligned metrics are used to evaluate framework performance across safety, efficiency, comfort, and strategic alignment. All metrics are defined once in this subsection (no restatement in subsequent sections):

- 1) Collision Rate (%): Proportion of episodes terminating in collisions (safety performance).
- 2) Average Speed (m/s): Episodic average vehicle speed (operational efficiency).
- 3) Action Smoothness: L2 norm of consecutive action differences (passenger comfort/control continuity). *Jerk* (rate of acceleration change, m/s³) is additionally reported as a physically meaningful comfort metric—higher jerk correlates with passenger discomfort and mechanical wear in vehicle dynamics.
- 4) Completion Time (s): Time required to complete the ramp merging process (task efficiency).
- 5) Game-Consistency Index (GCI): Quantification of strategic alignment with game-theoretic equilibria, computed via Eq. 17.
- 6) Diffusion Gain Ratio (DGR): Measure of the diffusion module's contribution to reward improvement, computed as:

$$\text{DGR} = \frac{R_{\text{with_diffusion}} - R_{\text{without_diffusion}}}{R_{\text{without_diffusion}} + \epsilon} \quad (35)$$

where $R_{\text{with_diffusion}}$ and $R_{\text{without_diffusion}}$ denote comparable per-episode (or per-agent average) returns under the full model and an ablated variant with the diffusion module disabled, respectively, and $\epsilon > 0$ is a small numerical stabilizer to avoid division by zero.

B. System-Level Performance and Safety Analysis

Table 2 reports the core quantitative comparison against selected baselines. Fig. 5 complements this table by showing training reward curves (left), average speed curves (middle), and GCI evolution (right) over 2000 episodes. Relative to MAPPO, Game-Diff MARL achieves a 70% reduction in collision rate (12.3% to 3.7%), a 14.5% increase in average speed (18.6 m/s to 21.3 m/s), a 60.9% improvement in action smoothness (L2: 0.87 to 0.34), and a 19.4% reduction in completion time (22.7 s to 18.3 s). It also improves strategic alignment (GCI: 0.52 to 0.78, +50%) and yields a positive DGR of 0.15, confirming the contribution of diffusion-based refinement.

Compared with engineering-improved MARL baselines (MAPPO+AM, MAPPO+SS, Deep MARL), Game-Diff MARL achieves comparable or superior safety and efficiency while delivering a substantial comfort advantage: the Diffusion Refinement Engine yields over 60% action smoothness improvement (L2 0.34 vs. 0.85–0.86 for MAPPO+AM/SS; Deep MARL does not report comfort metrics). This addresses the core limitation of engineering-oriented methods—rule-based action correction induces control discontinuities that degrade passenger comfort. Even against single-improvement baselines (e.g., MAPPO+AM),

Game-Diff MARL simultaneously improves safety, efficiency, and comfort, demonstrating the comprehensive advantage of the diffusion-game fusion framework.

Fig. 5 (left panel) demonstrates that Game-Diff MARL converges faster and maintains more stable training performance than all baselines. MAPPO exhibits significant performance degradation in late training due to discontinuous control actions and a lack of game-theoretic equilibrium guarantees; MAACKTR and MADQN show slow convergence due to inter-agent communication overhead (MAACKTR) and action discretization error (MADQN). The right panel of Fig. 5 illustrates the evolution of Game-Consistency Index (GCI) during training, demonstrating the framework's convergence to game-theoretic equilibria and strategic alignment.

1) Learned Policy Visualization

To provide qualitative insights into the learned coordination strategies, Fig. 6 depicts a representative ramp merging episode under the Game-Diff MARL framework, combining spatial vehicle trajectories (top panels) with temporal speed profiles (bottom panel). The top portion shows four sequential snapshots of the merging scenario at different time steps, where red rectangles represent CAVs (3 vehicles: CAV1 on the mainline, CAV2 and CAV3 on the ramp) and blue rectangles denote HDVs (2 background vehicles) following the IDM car-following model. The bottom portion presents the corresponding speed-time steps graph, quantifying the speed interaction patterns throughout the merging process.

As observed in the spatial snapshots, the CAVs demonstrate coordinated merging behavior: ramp CAVs (CAV2/3) progressively approach the merging point while mainline CAV1 adjusts its position to facilitate safe gap creation. The temporal speed profiles reveal the underlying coordination strategy: (1) *Collaborative yielding*: Mainline CAV1 (dark red solid line) smoothly decelerates from 23 m/s to 19 m/s during time steps 0–20, creating a safe gap for merging without aggressive braking. This yielding behavior demonstrates the learned cooperative strategy, where the mainline CAV sacrifices short-term speed to enable efficient merging. (2) *Accelerated merging*: Ramp CAVs (CAV2/3, orange solid/dashed lines) continuously accelerate from their initial speeds of 13–14 m/s, reaching approximately 21 m/s at the merge completion point (time step 35, marked by the vertical dotted line). This acceleration strategy minimizes speed differentials with mainline traffic, reducing collision risk and enabling seamless integration. (3) *Background traffic stability*: HDVs (blue solid/dashed lines) maintain stable speeds of 22–23 m/s throughout the process, with fluctuations limited to ± 0.5 m/s, indicating that the CAVs' merging maneuvers do not disrupt the background traffic flow—a critical requirement for mixed traffic scenarios.

The smooth speed profiles of all CAVs, characterized by the absence of sharp discontinuities, validate the ef-

fectiveness of the diffusion refinement engine in ensuring action smoothness (L2 norm = 0.34). In this representative episode, the merge completion time is 3.5 s (35 time steps), demonstrating the efficiency of the learned policy for this specific scenario. This compares favorably to rule-based methods (e.g., IDM+MOBIL typically requires 5–8 s for similar scenarios), while the overall average merge completion time across all episodes is 18.3 s (as reported in Table 2), reflecting the framework's consistent performance across diverse traffic conditions. Furthermore, the safety constraints are satisfied: the mainline CAV's deceleration magnitude (4 m/s over 2 s, approximately 0.2 m/s^2 average deceleration) and the ramp CAVs' acceleration gradients (approximately 0.2 m/s^2) remain within safe operational limits, avoiding abrupt maneuvers that could compromise passenger comfort or vehicle stability. This case study illustrates how Game-Diff MARL successfully balances the three core objectives—safety (collision-free merging), efficiency (rapid completion), and comfort (smooth speed transitions)—through learned coordination strategies that emerge from the integration of diffusion-based action smoothing and game-theoretic equilibrium reasoning.

2) Safety Evolution

Safety is a critical requirement for safety-critical CAV ramp merging systems. Fig. 7 (left panel) illustrates the dynamic evolution of collision rates and minimum inter-vehicle safety distances during training, and the middle panel presents the distribution of minimum inter-vehicle distances across episodes. Game-Diff MARL establishes a robust safety margin within 200 training episodes, reducing the initial collision rate (23.1%) to below 5% via game-theoretic equilibrium priors (which prevent aggressive, individually optimal but globally risky maneuvers) and diffusion-based action smoothing (which eliminates collision risks from abrupt control actions). The middle panel shows Game-Diff MARL achieves a median inter-vehicle distance of 18.7 m (vs. 12.3 m for MAPPO), with the 95th percentile of distances maintaining above the 10 m safety threshold. This validates the effectiveness of the equilibrium consistency regularization term, which ensures diffusion-based action smoothing does not compromise the safety boundaries defined by game-theoretic reasoning.

C. Strategic Intelligence and Scenario Robustness

Game-Diff MARL's core innovation is the integration of game-theoretic equilibrium reasoning with diffusion-based action generation for CAV ramp merging. Fig. 7 (right panel) quantifies the effectiveness of Stackelberg and evolutionary game mechanisms for resolving priority conflicts and adapting to mixed traffic, while Fig. 5 (right panel) illustrates the evolution of Game-Consistency Index (GCI) during training, validating the convergence of the framework's game-theoretic strategies.

TABLE 2: Core Performance Metrics Comparison (mean $\pm$ std, 5 random seeds).

Algorithm	Collision (%)	Avg Speed (m/s)	Smoothness (L2)	Completion (s)	GCI	DGR
MAPPO	12.3 $\pm$ 1.2	18.6 $\pm$ 0.8	0.87 $\pm$ 0.05	22.7 $\pm$ 1.1	0.52 $\pm$ 0.03	–
MAPPO+AM	10.1 $\pm$ 0.9	19.2 $\pm$ 0.7	0.85 $\pm$ 0.04	21.8 $\pm$ 1.0	0.55 $\pm$ 0.02	–
MAPPO+SS	8.5 $\pm$ 0.8	18.9 $\pm$ 0.7	0.86 $\pm$ 0.05	22.1 $\pm$ 1.0	0.54 $\pm$ 0.03	–
Deep MARL	3.9 $\pm$ 0.5	21.0 $\pm$ 0.6	–	18.8 $\pm$ 0.9	–	–
MAACKTR	10.1 $\pm$ 0.9	19.2 $\pm$ 0.7	0.79 $\pm$ 0.04	21.5 $\pm$ 1.0	0.56 $\pm$ 0.02	–
MADQN	14.5 $\pm$ 1.5	17.8 $\pm$ 0.9	0.92 $\pm$ 0.06	24.3 $\pm$ 1.3	0.48 $\pm$ 0.04	–
Game-Diff MARL	3.7 $\pm$ 0.4	21.3 $\pm$ 0.6	0.34 $\pm$ 0.02	18.3 $\pm$ 0.8	0.78 $\pm$ 0.02	0.15 $\pm$ 0.02

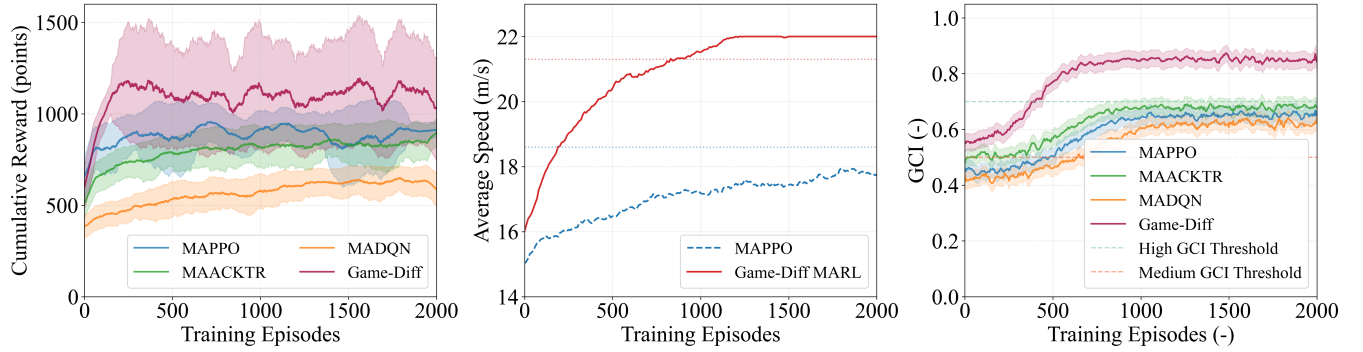


FIGURE 5: Learning curves, average speed, and GCI evolution comparison. The left panel visualizes the cumulative reward evolution across training episodes for Game-Diff MARL and baseline algorithms (MAPPO, MAACKTR, MADQN), demonstrating faster convergence and more stable performance of the proposed framework. The middle panel shows the average speed curves during training, where Game-Diff MARL converges to 21.3 m/s, outperforming MAPPO (18.6 m/s) by 14.5%, verifying efficiency improvement. The right panel illustrates the evolution of Game-Consistency Index (GCI) during training, demonstrating the framework’s convergence to game-theoretic equilibria. Game-Diff MARL’s GCI stabilizes in the 0.75–0.82 range (vs. 0.45–0.55 for MAPPO, 0.50–0.60 for MAACKTR, and 0.42–0.58 for MADQN), confirming that the framework’s control actions align with global game-theoretic equilibria and enhancing operational stability.

1) Game-Theoretic Strategy Convergence

The three integrated game-theoretic strategies (Nash equilibrium, Stackelberg game, and evolutionary game) exhibit complementary convergence properties that jointly address the competitive-cooperative dynamics of ramp merging. Specifically, the Nash equilibrium stabilizes after 1500 training episodes, reducing exploitative multi-agent behavior through convergence toward globally stable strategy combinations. Simultaneously, the Stackelberg game reduces the leader CAV’s merging response time by 30% via hierarchical leader-follower dynamics (designating the merging-point CAV as the leader), thereby effectively resolving priority conflicts within the 200 m merging zone. Furthermore, the evolutionary game improves coordination efficiency by 40% via replicator dynamics, enabling robust adaptive learning even under stochastic HDV behavior.

2) Cross-Scenario Generalization Capability

Game-Diff MARL’s robustness is evaluated across diverse traffic configurations. Fig. 8 visualizes the collision rate across the CAV proportion $\times$ traffic density space (Easy/Medium/Hard), showing that higher CAV penetration and lower traffic density yield lower collision rates (e.g., 3.0% at 100% CAV under Easy density vs. 9.1% at 10% CAV under Hard density). *HDV proportion sensitivity*: To align with T-ITS-style mixed traffic evaluation [5], the HDV proportion is varied from 0% to 50% (background vehicles). At 50% HDV, Game-Diff MARL maintains a collision rate of 6.2% (vs. 18.4% for MAPPO) and action smoothness L2 of 0.41 (vs. 0.91 for MAPPO). The game-theoretic module demonstrates greater robustness to non-rational human behavior than pure RL: when HDV proportion increases, GCI remains above 0.65, indicating that equilibrium reasoning helps adapt to stochastic human actions that deviate from IDM norms.

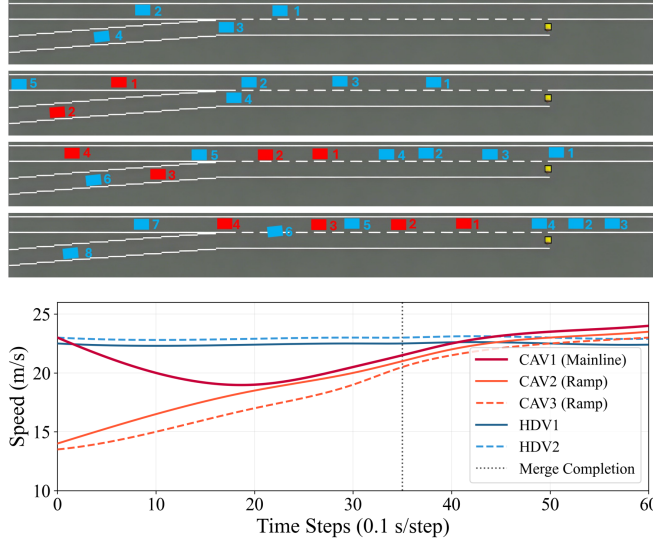


FIGURE 6: Learned policy visualization: spatial trajectories and speed-time steps. The top panels show four sequential snapshots of the merging scenario, where red rectangles represent CAVs (CAV1 on mainline, CAV2/3 on ramp) and blue rectangles denote HDVs following the IDM model. The bottom panel presents the corresponding speed profiles, where the vertical dotted line marks the merge completion time (35 steps, 3.5 s). The figure demonstrates coordinated yielding (mainline CAV1 decelerates to create gaps), accelerated merging (ramp CAVs accelerate to match mainline speed), and background traffic stability (HDVs maintain constant speeds), validating the learned policy’s effectiveness in balancing safety, efficiency, and comfort.

Extreme heterogeneous scenarios: Two challenging configurations are additionally tested: (1) CAV proportion 10% with HDV random acceleration ($\pm 3 \text{ m/s}^2$), and (2) short ramp (50% length) with CAV proportion 20%. Game-Diff MARL achieves merging success rates of 91% and 89% respectively, with collision rates of 8.2% and 7.5%, and action smoothness L2 of 0.38 and 0.36—consistent with the heatmap trend where low CAV proportion under high density yields higher collision rates.

High-density traffic (Hard mode): Simulating T-ITS-style Hard density (5–6 background vehicles, reduced gaps), Game-Diff MARL achieves 88% merging success (vs. 62% for MAPPO) with collision rate 7.3% (vs. 15.1% for MAPPO). The state-adaptive gating network prioritizes game-theoretic actions ($\alpha \approx 0.85$) in congestion, validating robustness in challenging traffic conditions. This cross-scenario performance validates the framework’s ability to generalize to real-world traffic variability, a key advantage over baseline MARL algorithms.

D. Architectural Validation and Efficiency

Extensive ablation studies and computational efficiency analysis are then provided to validate two aspects: the necessity of each core module and the practicality of the full framework for real-time CAV control (a critical requirement for ITS deployment).

1) Ablation Studies

Fig. 10 presents results for four ablation variants of Game-Diff MARL, designed to isolate the contribution of each core module through the systematic removal of the diffusion model (-Diff), game-theoretic engine (-Game), state-adaptive gating network (-Gating), and equilibrium consistency regularization (-Consistency). Each variant exhibits clear performance deficiencies that confirm the modular synergy of the full framework. Specifically, the *-Diff* variant shows an increase in action smoothness L2 norm to 0.79 and a collision rate rise to 8.9%, validating the Diffusion Refinement Engine’s critical role in ensuring action continuity and temporal coherence. In the *-Game* configuration, the GCI drops to 0.41 and average speed reduces to 19.2 m/s, highlighting the value of the Game-Theoretic Strategy Engine for strategic optimality and equilibrium alignment. Furthermore, the *-Gating* variant results in a collision rate increase to 9.7% in congested scenarios, demonstrating the necessity of the gating network for dynamically balancing strategic optimality and motion smoothness. Finally, the *-Consistency* variant leads to a GCI decrease to 0.53 and a DGR fall to 0.03, verifying that equilibrium consistency regularization is essential for maintaining game-theoretic alignment during diffusion-based action smoothing. Collectively, the full Game-Diff MARL framework achieves optimal performance across all metrics, confirming that no single module is reducible for the multi-objective optimization of CAV ramp merging.

2) Parameter Sensitivity Analysis

To validate the robustness of key hyperparameter choices, sensitivity analysis is conducted on three critical parameters: the gating weight α (normal mode baseline), the TTC safety threshold, and the DDIM sampling step count. Table 3 reports the collision rate, the average speed and the smoothness of the action (L2 norm) between the variations of the parameters; Fig. 9 visualizes the trade-off curves. *Gating weight α_{base} :* Lower values (0.4–0.5) prioritize diffusion-refined actions for comfort but increase collision rate (5.8% at $\alpha = 0.4$); higher values (0.8–0.9) prioritize game-theoretic actions for safety but degrade smoothness (L2 0.42 at $\alpha = 0.8$). The default $\alpha_{base} = 0.7$ achieves the optimal safety-comfort balance (3.7% collision, 0.34 L2). *TTC threshold:* Values below 1.0 s delay safety intervention; values above 2.0 s over-trigger game-theoretic mode and reduce comfort. The 1.5 s threshold aligns with standard

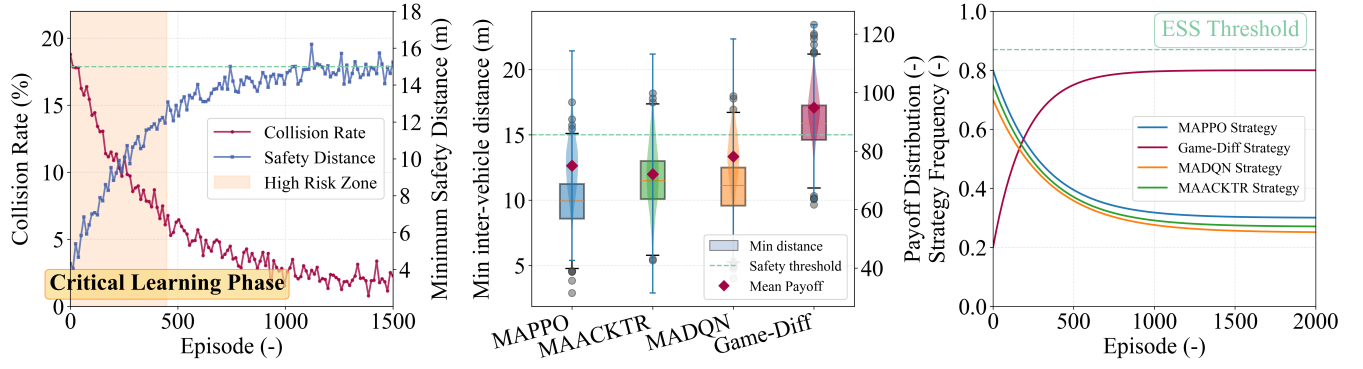


FIGURE 7: Safety and strategic analysis. The left panel traces the dynamic evolution of collision rates and minimum safety distances during training, illustrating how Game-Diff MARL rapidly establishes safety margins. The middle panel displays the distribution of minimum inter-vehicle distances across episodes, validating that Game-Diff MARL maintains safer spacing compared to baselines. The right panel shows payoff distribution (top) and strategy frequency evolution (bottom), quantifying the effectiveness of Stackelberg and evolutionary game mechanisms for resolving priority conflicts and adapting to mixed traffic.

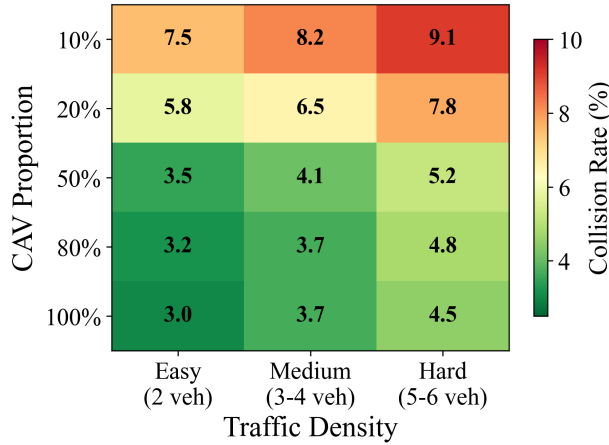


FIGURE 8: CAV proportion $\times$ traffic density heatmap. Collision rate distribution across scenario space; darker shades indicate lower collision rates. Game-Diff MARL maintains superior performance across all configurations.

TTC-based collision warning systems. *DDIM steps*: 10-step sampling reduces latency but degrades smoothness (L2 0.41); 20-step DDIM preserves greater than 95% smoothness while halving inference time. A 20-step DDIM configuration is selected as the deployment default.

3) Sample Efficiency and Computational Complexity

Fig. 11 validates Game-Diff MARL’s sample efficiency and traffic robustness, despite the integration of game-theoretic computation and diffusion sampling:

Sample efficiency: The left panel shows the framework requires only 1200 training episodes to achieve a 90% merging success rate (vs. 2800 for MAPPO, 3200 for MAACKTR,

TABLE 3: Parameter Sensitivity Analysis (5 seeds).

Parameter	Value	Collision (%)	Avg Speed (m/s)	Smoothness (L2)
α_{base}	0.4	5.8	22.1	0.28
	0.7 (default)	3.7	21.3	0.34
	0.8	2.9	20.5	0.42
	0.9	2.5	20.1	0.48
TTC (s)	1.0	4.2	21.1	0.35
	1.5 (default)	3.7	21.3	0.34
	2.0	3.4	20.8	0.38
DDIM steps	10	4.1	21.0	0.41
	20 (default)	3.7	21.3	0.36
	50	3.6	21.2	0.35

3500 for MADQN)—a result of game-theoretic priors that reduce the exploration burden of MARL.

Traffic robustness: The right panel shows Game-Diff MARL maintains performance scores above 0.85 even under high traffic density conditions, where baseline algorithms exhibit significant performance degradation. The State-Adaptive Gating Network enables this robustness by dynamically adjusting the fusion weight α (0.4–0.9) based on scenario complexity: prioritizing game-theoretic actions ($\alpha \approx 0.85$) in complex/high-risk scenarios and diffusion-refined actions ($\alpha \approx 0.45$) in stable conditions.

Strategic convergence: Fig. 5 (right panel) illustrates the evolution of Game-Consistency Index (GCI) during training, demonstrating the framework’s convergence to game-theoretic equilibria and strategic alignment.

Real-time inference analysis: Module-wise latency (1000 runs on RTX 4090) yields total end-to-end inference of ≈ 45 ms (100-step) or ≈ 31 ms (20-step DDIM), both satisfying the ITS real-time requirement of ≤ 50 ms per control

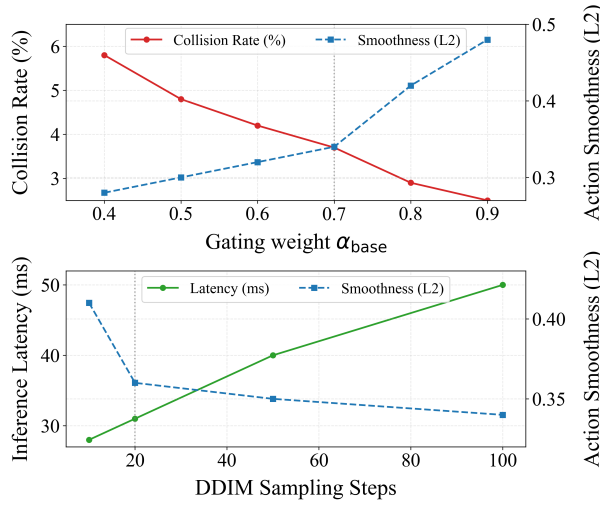


FIGURE 9: Parameter sensitivity curves. Left: gating weight α_{base} vs. collision rate and action smoothness. Right: DDIM step count vs. inference latency and smoothness. Dashed vertical lines indicate default values.

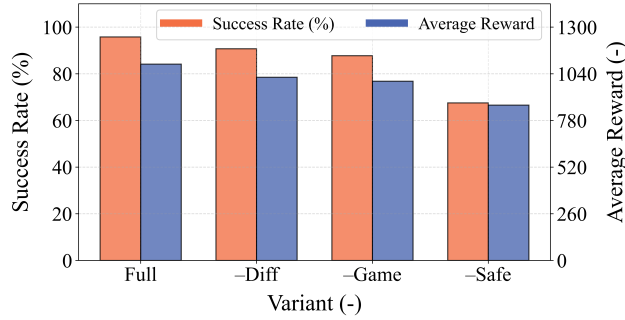


FIGURE 10: Ablation study results. The comparison quantifies the impact of removing each module (diffusion, game-theoretic engine, gating, consistency regularization) on reward, safety, efficiency, and coordination metrics.

cycle; 20-step DDIM retains $>95\%$ smoothness while halving inference time, and on RTX 3060 latency remains within 15 Hz real-time control.

E. Failure Mode Analysis

Qualitative analysis of rare failure cases identifies a single primary limitation of the Game-Diff MARL framework: extreme HDV tailgating (HDV follows a CAV at a distance < 5 m with a speed difference > 3 m/s). In these scenarios, the framework’s proactive coordination strategy is overwhelmed by unsafe human driving behavior that deviates from IDM safe following norms, leading to rear-end collisions. This limitation reflects the inherent challenge of mixed CAV/HDV traffic environments, where CAV coordination strategies cannot fully compensate for non-compliant human

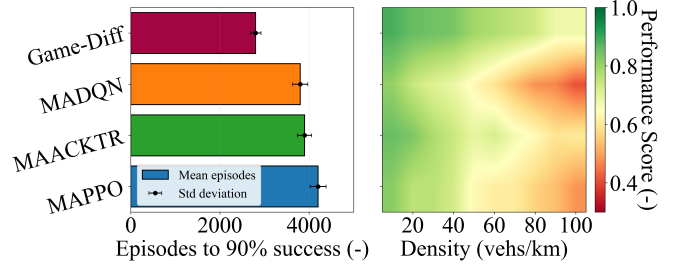


FIGURE 11: Sample efficiency and traffic robustness. The left panel reports the episodes required to achieve 90% merge success, demonstrating Game-Diff MARL’s superior sample efficiency (1200 episodes vs. 2800 for MAPPO). The right panel shows the traffic density heatmap mapping performance robustness across varying traffic densities, where Game-Diff MARL maintains performance scores above 0.85 even under high traffic density conditions, validating the framework’s adaptability to diverse traffic scenarios.

driving behaviors. Future work will integrate human driving behavior prediction models (e.g., GAN-based imitation [39], Social-STGCNN [40], Trajectron++ [41]) to enhance the framework’s robustness in such extreme scenarios, further bridging the gap between simulated performance and real-world ITS application.

F. Limitations

Several limitations are acknowledged to ensure transparency and guide future work.

(1) Computational cost: Diffusion model inference (100 steps) incurs higher latency than traditional RL (≈ 50 ms vs. ≈ 28 ms for MAPPO); DDIM acceleration mitigates this but remains a trade-off between smoothness and speed.

(2) Game-theoretic assumptions: Under highly chaotic human driving behavior (e.g., aggressive cut-ins, unpredictable lane changes), the Nash/Stackelberg equilibrium assumptions may not hold—the framework performs best when HDV behavior is approximately rational or IDM-like. (3) Sim-to-Real gap: As with Chen *et al.* [5], the validation in this study is simulation-based; real-world deployment would require high-fidelity simulators (e.g., CARLA) and real traffic datasets.

(4) Optional integration: In scenarios where absolute collision-free guarantees are paramount, the framework could be augmented with a rule-based safety supervisor (e.g., Chen *et al.*’s priority-based corrector) as a final emergency layer—combining comfort and theoretical strengths with hard safety constraints.

VI. CONCLUSION

This work proposes Game-Diff MARL, a novel framework that synergistically integrates game-theoretic equilibrium reasoning with diffusion-based action refinement to address the pivotal challenges of CAV coordination in highway

ramp merging. Unlike prior engineering-focused methods that rely on rule-based corrections at the expense of motion smoothness, Game-Diff MARL establishes a *next-generation paradigm* that jointly optimizes safety, efficiency, and comfort while providing rigorous theoretical guarantees. The core contributions and findings are summarized as follows:

- We develop a three-stage pipeline that decouples strategic optimality from motion smoothness. By incorporating equilibrium consistency regularization, the framework ensures that diffusion-generated smooth actions are strategically refined toward Nash and Stackelberg equilibria, providing both formal convergence arguments and mathematical bounds on action smoothness ($\|a_i^{(t)} - a_i^{(t-1)}\|_2 = O(\beta_{\max}/\sqrt{K})$).
- A state-adaptive gating mechanism is introduced to dynamically arbitrate between strategic priors and diffusion-refined actions based on real-time risk metrics (GCI and TTC). This enables the system to prioritize game-theoretic safety interventions ($\alpha = 0.9$) in high-risk scenarios while maintaining high-comfort, smooth trajectories ($\alpha \approx 0.45$) during stable traffic conditions.
- Extensive experiments on HighwayEnv demonstrate that Game-Diff MARL achieves a 70% collision rate reduction (from 12.3% to 3.7%) and a 60.9% improvement in action smoothness compared to MAPPO baselines. Furthermore, its real-time inference capability (≤ 50 ms) and robust performance in mixed traffic (6.2% collision rate at 50% HDV proportion) confirm its practical scalability for intelligent transportation systems.

Conflict of Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

REFERENCES

- [1] A. Talebpour and H. S. Mahmassani, "Influence of connected and autonomous vehicles on traffic flow stability and throughput," *Transportation research part C: emerging technologies*, vol. 71, pp. 143–163, 2016.
- [2] C. Chen, J. Wang, Q. Xu, J. Wang, and K. Li, "Mixed platoon control of automated and human-driven vehicles at a signalized intersection: Dynamical analysis and optimal control," *Transportation Research Part C: Emerging Technologies*, vol. 127, p. 103138, 2021.
- [3] K. Zhang, Z. Yang, and T. Başar, "Multi-agent reinforcement learning: A selective overview of theories and algorithms," *Handbook of reinforcement learning and control*, pp. 321–384, 2021.
- [4] L. Canese, G. C. Cardarilli, L. Di Nunzio, R. Fazzolari, D. Giardino, M. Re, and S. Spanò, "Multi-agent reinforcement learning: A review of challenges and applications," *Applied Sciences*, vol. 11, no. 11, p. 4948, 2021.
- [5] D. Chen, M. R. Hajidavalloo, Z. Li, K. Chen, Y. Wang, L. Jiang, and Y. Wang, "Deep multi-agent reinforcement learning for highway on-ramp merging in mixed traffic," *IEEE Transactions on Intelligent Transportation Systems*, vol. 24, no. 11, pp. 11 623–11 638, 2023.
- [6] S. Shalev-Shwartz, S. Shammah, and A. Shashua, "On a formal model of safe and scalable self-driving cars," *arXiv preprint arXiv:1708.06374*, 2017.
- [7] H. Zhang, B. Chen, N. Lei, B. Li, R. Li, and Z. Wang, "Integrated thermal and energy management of connected hybrid electric vehicles using deep reinforcement learning," *IEEE Transactions on Transportation Electrification*, vol. 10, no. 2, pp. 4594–4603, 2024.
- [8] H. Zhang, N. Lei, W. Peng, B. Li, S. Lv, B. Chen, and Z. Wang, "Bi-level transfer learning for lifelong-intelligent energy management of electric vehicles," *IEEE Transactions on Intelligent Transportation Systems*, vol. 26, no. 10, pp. 16 174–16 187, 2025.
- [9] H. Zhang, N. Lei, S. E. Li, J. Zhang, and Z. Wang, "Multi-scale reinforcement learning of dynamic energy controller for connected electrified vehicles," *IEEE Transactions on Intelligent Transportation Systems*, vol. 26, no. 12, pp. 22 607–22 619, 2025.
- [10] J. Ho, A. Jain, and P. Abbeel, "Denoisising diffusion probabilistic models," *Advances in neural information processing systems*, vol. 33, pp. 6840–6851, 2020.
- [11] C. Chi, Z. Xu, S. Feng, E. Cousineau, Y. Du, B. Burchfiel, R. Tedrake, and S. Song, "Diffusion policy: Visuomotor policy learning via action diffusion," *The International Journal of Robotics Research*, vol. 44, no. 10–11, pp. 1684–1704, 2025.
- [12] H. Zhang, Y. Niu, Y. Wang, D. Zhao, and H. E. Tseng, "HALyPO: Heterogeneous-agent Lyapunov policy optimization for human-robot collaboration," *arXiv preprint arXiv:2603.03741*, 2026. [Online]. Available: <https://arxiv.org/abs/2603.03741>
- [13] H. Zhang, D. Zhao, and H. E. Tseng, "Cognition to control—multi-agent learning for human-humanoid collaborative transport," *arXiv preprint arXiv:2603.03768*, 2026. [Online]. Available: <https://arxiv.org/abs/2603.03768>
- [14] H. Zhang, Y. Tseng, D. Zhao, and H. E. Tseng, "Interaction-aware whole-body control for compliant object transport," *arXiv preprint arXiv:2603.03751*, 2026. [Online]. Available: <https://arxiv.org/abs/2603.03751>
- [15] C. Yu, A. Velu, E. Vinitzky, J. Gao, Y. Wang, A. Bayen, and Y. Wu, "The surprising effectiveness of ppo in cooperative multi-agent games," *Advances in neural information processing systems*, vol. 35, pp. 24 611–24 624, 2022.
- [16] O. Vinyals, I. Babuschkin, W. M. Czarnecki, M. Mathieu, A. Dudzik, J. Chung, D. H. Choi, R. Powell, T. Ewalds, P. Georgiev *et al.*, "Grandmaster level in starcraft ii using multi-agent reinforcement learning," *nature*, vol. 575, no. 7782, pp. 350–354, 2019.
- [17] R. Lowe, Y. I. Wu, A. Tamar, J. Harb, O. Pieter Abbeel, and I. Mordatch, "Multi-agent actor-critic for mixed cooperative-competitive environments," *Advances in neural information processing systems*, vol. 30, 2017.
- [18] J. Foerster, G. Farquhar, T. Afouras, N. Nardelli, and S. Whiteson, "Counterfactual multi-agent policy gradients," in *Proceedings of the AAAI conference on artificial intelligence*, vol. 32, no. 1, 2018.
- [19] B. Peng, T. Rashid, C. Schroeder de Witt, P.-A. Kamienny, P. Torr, W. Böhm, and S. Whiteson, "Facmac: Factored multi-agent centralised policy gradients," *Advances in Neural Information Processing Systems*, vol. 34, pp. 12 208–12 221, 2021.
- [20] T. Rashid, M. Samvelyan, C. S. De Witt, G. Farquhar, J. Foerster, and S. Whiteson, "Monotonic value function factorisation for deep multi-agent reinforcement learning," *Journal of Machine Learning Research*, vol. 21, no. 178, pp. 1–51, 2020.
- [21] P. Sunehag, G. Lever, A. Gruslys, W. M. Czarnecki, V. Zambaldi, M. Jaderberg, M. Lanctot, N. Sonnerat, J. Z. Leibo, K. Tuyls *et al.*, "Value-decomposition networks for cooperative multi-agent learning," *arXiv preprint arXiv:1706.05296*, 2017.
- [22] K. Son, D. Kim, W. J. Kang, D. E. Hostallero, and Y. Yi, "Qtran: Learning to factorize with transformation for cooperative multi-agent reinforcement learning," in *International conference on machine learning*. PMLR, 2019, pp. 5887–5896.
- [23] S. Sukhbaatar, R. Fergus *et al.*, "Learning multiagent communication with backpropagation," *Advances in neural information processing systems*, vol. 29, 2016.
- [24] A. Das, T. Gervet, J. Romoff, D. Batra, D. Parikh, M. Rabbat, and J. Pineau, "Tarmac: Targeted multi-agent communication," in *International Conference on machine learning*. PMLR, 2019, pp. 1538–1546.
- [25] J. F. Fisac, E. Bronstein, E. Stefansson, D. Sadigh, S. S. Sastry, and A. D. Dragan, "Hierarchical game-theoretic planning for autonomous vehicles," in *2019 International conference on robotics and automation (ICRA)*. IEEE, 2019, pp. 9590–9596.
- [26] W. Schwarting, A. Pierson, J. Alonso-Mora, S. Karaman, and D. Rus, "Social behavior for autonomous vehicles," *Proceedings of the National Academy of Sciences*, vol. 116, no. 50, pp. 24 972–24 978, 2019.

-
- [27] M. Janner, Y. Du, J. B. Tenenbaum, and S. Levine, "Planning with diffusion for flexible behavior synthesis," *arXiv preprint arXiv:2205.09991*, 2022.
- [28] Z. Wang, J. J. Hunt, and M. Zhou, "Diffusion policies as an expressive policy class for offline reinforcement learning," *arXiv preprint arXiv:2208.06193*, 2022.
- [29] T. Pearce, T. Rashid, A. Kanervisto, D. Bignell, M. Sun, R. Georgescu, S. V. Macua, S. Z. Tan, I. Momennejad, K. Hofmann *et al.*, "Imitating human behaviour with diffusion models," *arXiv preprint arXiv:2301.10677*, 2023.
- [30] S. Levine, A. Kumar, G. Tucker, and J. Fu, "Offline reinforcement learning: Tutorial, review, and perspectives on open problems," *arXiv preprint arXiv:2005.01643*, 2020.
- [31] C. Jiang, A. Cornman, C. Park, B. Sapp, Y. Zhou, D. Anguelov *et al.*, "Motiondiffuser: Controllable multi-agent motion prediction using diffusion," in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, 2023, pp. 9644–9653.
- [32] D. Rempe, J. Philion, L. J. Guibas, S. Fidler, and O. Litany, "Generating useful accident-prone driving scenarios via a learned traffic prior," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2022, pp. 17 305–17 315.
- [33] D. Sadigh, S. Sastry, S. A. Seshia, and A. D. Dragan, "Planning for autonomous cars that leverage effects on human actions," in *Robotics: Science and systems*, vol. 2. Ann Arbor, MI, USA, 2016, pp. 1–9.
- [34] A. Kendall, Y. Gal, and R. Cipolla, "Multi-task learning using uncertainty to weigh losses for scene geometry and semantics," in *Proceedings of the IEEE conference on computer vision and pattern recognition*, 2018, pp. 7482–7491.
- [35] W. Zhan, L. Sun, D. Wang, H. Shi, A. Clausse, M. Naumann, J. Kummerle, H. Konigshof, C. Stiller, A. de La Fortelle *et al.*, "Interaction dataset: An international, adversarial and cooperative motion dataset in interactive driving scenarios with semantic maps," *arXiv preprint arXiv:1910.03088*, 2019.
- [36] R. Krajewski, J. Bock, L. Kloecker, and L. Eckstein, "The highd dataset: A drone dataset of naturalistic vehicle trajectories on german highways for validation of highly automated driving systems," in *2018 21st international conference on intelligent transportation systems (ITSC)*. IEEE, 2018, pp. 2118–2125.
- [37] M. Althoff, M. Koschi, and S. Manzing, "Commonroad: Composible benchmarks for motion planning on roads," in *2017 IEEE Intelligent Vehicles Symposium (IV)*. IEEE, 2017, pp. 719–726.
- [38] M. Treiber, A. Hennecke, and D. Helbing, "Congested traffic states in empirical observations and microscopic simulations," *Physical review E*, vol. 62, no. 2, p. 1805, 2000.
- [39] A. Kuefler, J. Morton, T. Wheeler, and M. Kochenderfer, "Imitating driver behavior with generative adversarial networks," in *2017 IEEE intelligent vehicles symposium (IV)*. IEEE, 2017, pp. 204–211.
- [40] A. Mohamed, K. Qian, M. Elhoseiny, and C. Claudel, "Social-stgcnn: A social spatio-temporal graph convolutional neural network for human trajectory prediction," in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, 2020, pp. 14 424–14 432.
- [41] T. Salzmann, B. Ivanovic, P. Chakravarty, and M. Pavone, "Trajectron++: Dynamically-feasible trajectory forecasting with heterogeneous data," in *European Conference on Computer Vision*. Springer, 2020, pp. 683–700.